1 A student is investigating the resistance of a light-dependent resistor (LDR) separated from a source of light by different depths of water as shown in Fig. 1.1.

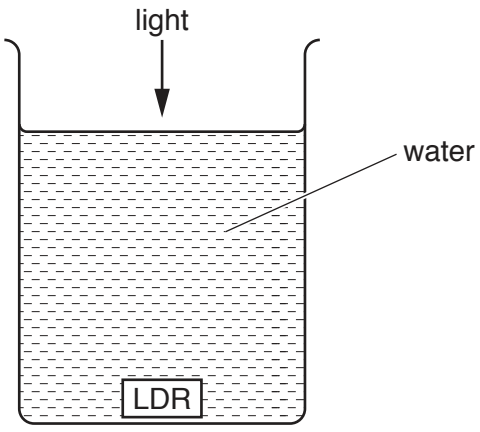


Fig. 1.1

It is suggested that the relationship between the resistance R of the LDR and the depth d of the LDR in the water is

$$R = \frac{4\pi d^2}{K}$$

where K is a constant.

Design a laboratory experiment to test the relationship between R and d. Explain how your results could be used to determine a value for K. You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

- the procedure to be followed,
- the measurements to be taken,
- the control of variables,
- the analysis of the data,
- any safety precautions to be taken.

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